DANIEL KARIMI

TELECOMMUNICATIONS ENGINEER · SIGNAL PROCESSING

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Talent Attraction Team – Indra Group
Defence and Security Division
Madrid, Spain

RE: Application for Telecommunications Engineer - Waveform & Data Link Design

Dear Talent Attraction Team at Indra Group,

I am writing to express my strong interest in the **Telecommunications Engineer - Waveform & Data Link Design** position within the Defence and Security division. As a soon-to-be graduate in **Telecommunications Systems Engineering at Universidad Politécnica de Madrid (ETSIT-UPM)**, I have cultivated a deep academic and practical foundation in Digital Signal Processing (DSP), modern modulation techniques, and channel coding—the core competencies demanded for Layer 1 and Layer 2 communication system design.

Throughout my degree, I gained extensive hands-on experience using **MATLAB and Simulink** to model and validate communication workflows. My academic projects extensively covered the design and simulation of waveforms (including QAM and OFDM), digital filters (FIR/IIR), error correction protocols (convolutional codes), and the precise analysis of bit error rates (BER) under simulated noisy conditions such as AWGN and fading environments. Furthermore, my laboratory courses established a robust understanding of digital hardware integration, giving me practical exposure to **VHDL and the fundamentals of FPGA architectures**, critical for bridging the gap between theoretical designs and physical implementations.

Why my profile aligns with the role:
· Strong foundation in OSI Layers 1 and 2, DSP techniques, and signal synchronization.
· Demonstrated proficiency in MATLAB/Simulink modeling and validation.
· Experience with large-scale signal processing and system simulation acquired during my CSIC research internship.
· Strong programming logic (C/C++, Python) combined with a solid foundation in digital logic systems and VHDL.

During my recent internship at the **Spanish National Research Council (CSIC-ITEFI)**, I further developed my system-level engineering skills. I was tasked with analyzing large-scale temporal data and implementing robust DSP algorithms in Python, validating their behavior through customized simulation environments—a workflow directly comparable to the model-based design and validation of tactical communication systems. This experience reinforced my capability to contribute to the entire development lifecycle, from architectural design to integration and rigorous testing.

The prospect of joining Indra's Defence division to work on mission-critical communication protocols and optimize waveforms on RFSoC/MPSoC solutions is an incredibly compelling technical challenge. I am eager to leverage my fresh academic knowledge and research-backed problem-solving approach to bring value to your engineering team.

Please find attached my resume and a recommendation letter from my CSIC supervisors detailing my analytical capabilities and dedication in complex technical environments. I kindly welcome the opportunity to discuss how my qualifications align with Indra's innovative defense projects.

Sincerely,

DANIEL KARIMI ÁLVAREZ
BSC TELECOMMUNICATIONS SYSTEMS ENGINEERING · UPM